



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
MONDAY TEST

CLASS: IX
SUBJECT: MATHEMATICS

FULL MARKS: 40
DATE: 19.06.23

Instructions:

- The paper consists of two printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

SECTION: A

1. Choose the correct option

[1 × 5 = 5]

i) If $x + \frac{1}{x} = 3$; then the value of $x^2 + \frac{1}{x^2}$ is:

- a) 3 b) 12 c) 11 d) 7

ii) Two rational numbers between $\frac{3}{7}$ and $\frac{4}{7}$ are:

- a) $\frac{1}{2}, \frac{18}{35}$ b) $\frac{41}{70}, \frac{2}{3}$ c) $\frac{5}{14}, \frac{31}{70}$ d) $\frac{9}{14}, \frac{43}{70}$

iii) The term to be added to $4p^2 + 25q^2 - 16pq$ to make it a perfect square trinomial is:

- a) $-4pq$ b) $12pq$ c) $-2pq$ d) $4pq$

iv) The factors of $16(x + y)^2 - 9(x - y)^2$ are:

- a) $(x + 7y)(x - 7y)$ b) $(x + 5y)(x - 5y)$ c) $(x + 7y)(7x + y)$ d) $(x + 7y)(y - 7x)$

v) Every rational number is:

- a) Whole number b) Natural number c) Integer d) Real number

SECTION: B

2. Factorise:

[3+3]

i) $5a^4 - 5a^3 + 30a^2 - 30a$

ii) $2a^2b^2 + 3ab - 9$

3. By using suitable identities, evaluate the following: $(10.1)^3$

[3]

4. Factorise: $3 - 5x + 5y - 12(x - y)^2$

[3]

5. Convert 2.2565656..... into a rational number.

[3]

6. The difference between C.I. and S.I. on a certain sum of money at 10% per annum for 3 years is 620. Find the principal if the interest is compounded annually. [5]

7. If $a = \frac{1}{a-5}$, find i) $a - \frac{1}{a}$ ii) $a^2 - \frac{1}{a^2}$ [5]

8. Find the amount on 25000 for 3 years at 12% p.a, compounded annually and also find C.I after 3 years. [5]

9. The present population of the town is 15625. If the population increases 4% in the 1st year and then decreases 4% in the 2nd and 3rd year respectively, find the population of the town at the end of the 3 years. [5]